

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 7

(1) Research Program/Project/Study Title: Knowledge and Perspective of Students Towards Biodiversity and its Conservation and Protection

Proponent Name:	Lecel O. Bobier
Academic Rank:	Associate Professor V
Co-Authors (with academic rank):	Dr. Adelfo Z. Virtudazo -Prof. 6 Dr. Marife R. Hilapad -Asso. Prof. 5 Dr. Rhodora L. Ticar – Asso. Prof. 5 Dr. Lillibeth G. Leonor – Asso. Prof. 5 Prof. Virginia P. Cabales – Asso. Prof. 5 Prof. Merlyn F. Filosofo – Asso. Prof. 5 Prof. Clyde Dwyndel G. Leccio – Instructor 1
Campus/Satellite College/Address:	Mambusao Satellite College
Telephone/Cell Phone/Fax Number:	09466299997
E-mail Address:	lobobier@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation -Mambusao satellite College									
		Municipality/City:	Mambusao						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity		(8) Discipline/Field of Specialization Science -Biology							

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 2 of 7

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The biodiversity of life on earth is the foundation of human existence and well-being. Biodiversity and viable ecosystems protect us from natural disasters, regulate climate change, and provide food, fertile soil, and medicine. But biodiversity is declining and many of the biodiversity hotspots are located in countries burdened by poverty, food insecurity, and intensifying climate change. Each country should involve its people in protecting the environment and natural resources (Raga, 2013). Government and non-government are encouraged to sustain and save the environment preventing drastic and worse resource deterioration (Basir & Ming, 2019).

Unfortunately, both the world's forests and biodiversity are generally decreasing at an unpleasant rate that dictates the call for immediate conservation and protection of these resources (Coracero & Malabrigo, 2020). Conservation of nature is highly important for the good development of social and economic life (Morar and Peterlicean, 2012).

The knowledge from education are important tool in achieving conservation (Howe, 2019). Education can help people of various ages and social standing to find values, motivation skills, and responsibility regarding maintaining the environment and human health (Morar and Peterlicean, 2012).

The present generation, especially the children and youth are the two most significant actors that can prevent the total loss of biodiversity and natural resources (Morar & Peterlicean, 2012). Educating them at an early age can help improve their perspective and be fighters for biodiversity through active involvement in the conservation and protection of natural resources (Šorytė & Pakalniškienė, 2019).

This study will be conducted to determine the extent of the student's knowledge of biodiversity and understand their perspective on biodiversity conservation and protection since the three courses offered in this college are not science-related. The result of this study will be the basis for the development of instructional materials that can be used in the Science Technology and Society and Environmental Management in the Food Industry subject.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

1. to measure the student's level of knowledge on biodiversity;
2. to identify the perspective of students towards biodiversity;
3. to determine the relationship between student profile and their knowledge and perspective on biodiversity;
4. to determine the correlation between students' knowledge of biodiversity and their perspective towards its conservation and protection; and
5. to assess their knowledge and perspective on Philippine biodiversity laws and policies.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

Biodiversity and Its Importance

Biodiversity plays an important role in the presence, multiplication, and existence of the biological species in nature which are its important components and key elements for a better and healthy environment. It also plays a role in the conservation of natural resources (Patel, 2014).

Education plays an important role in biodiversity conservation. It is an important factor in shaping peoples' knowledge and attitudes toward biodiversity conservation; and hence should be a part of the curriculum in schools (Kideghesho, Roskaft, and Kaltenborn 2007). Educating students can provide a chance to improve the perspective and value allotment of students by improving their knowledge as studies suggest (Venuste et. al, 2017; Lindemann-Matthies et. al, 2014; Nyberg et, al, 2020).

In the study of Erhabor & Don, (2016) being knowledgeable about the environmental laws and principles of natural systems can help develop practical skills and improve students' ability to assess and prioritize the welfare and status of the environment.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 3 of 7

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

This study is anchored on the theory of *constructivism* in which educators can help their students to learn. It is based on the idea that people actively construct or make their own knowledge, and that reality is determined by their experiences as learner. Basically, learners use their previous knowledge as a foundation and build on it with new things that they learn. Students can learn more about biodiversity and developed a positive perspective towards biodiversity conservation and protection.

Behaviorism. The basic principles of this theory are relevant to instructional design. It can be used as basis for designing visual materials and gives rise to many related teaching strategies. Specific principles of this theory have direct relevance to instructional design and include: emphasis on producing observable and measurable outcomes in students; Pre-assessment of students to determine where instruction should begin; emphasis on mastering early steps before progressing to more complex levels of performance; use of reinforcement to input performance; and use of cues, shaping, and practice to ensure a strong stimulus-response association. These principles can be applied in developing instructional materials to address the gap of the learners.

Instructional System Development (ISD) is a system approach to analyzing, designing, implementing, and evaluating any instructional experience (Hodell, 2007; Clark, 2014). It is a set of procedures for systemically designing and developing an instructional material emphasizing primarily on what to do rather than on how to do it, or why it works and involves five basic phases: analysis, design, development, implementation, and evaluation which nest within it the Instructional Design Theory; a design-oriented or goal-oriented theory, an assumption that offers explicit guidance on how to better help people learn and develop. It aims for a learner-centered rather than the traditional-centered approach to instruction, so that effective learning can take place. This means that every component of the instruction is governed by the learning outcomes, which have been determined after a thorough analysis of the learners' need. The kinds of learning and development may include cognitive, emotional, social, physical, and spiritual aspects (Reigeluth, 2011; Clark, 2014). ISD is considered as a road map that helps ensure the learners achieve their learning and performance goals through formal, and informal environments (Hodell, 2007; Clark, 2014).

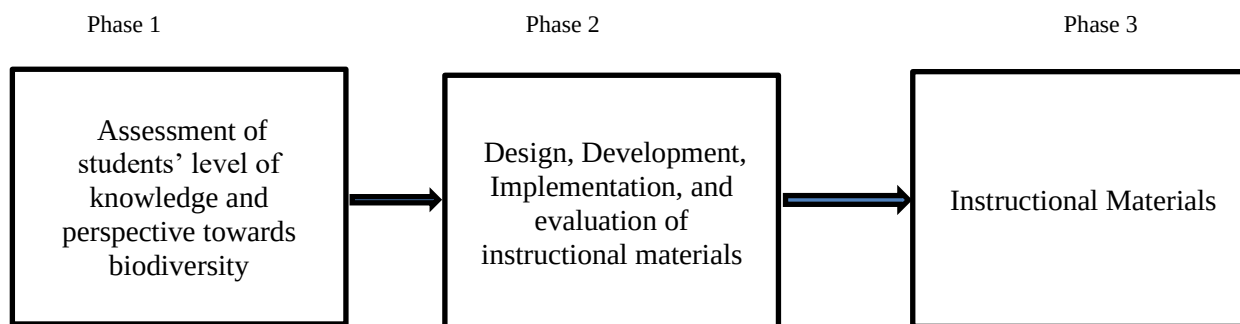


Figure 1. Conceptual framework showing how the instructional material will be developed based on the Level of students' knowledge and perspective towards biodiversity.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 4 of 7

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Research Design

The mixed-method research design will be used to analyze the students' responses. This design is efficient in using both qualitative and quantitative methods to assert meanings by using words and numbers (Almeida et al., 2016)

Research Methodology

The adopted survey questionnaire will be used in this study. It is composed of four parts;

Part I – Student's profile

Part II – Knowledge of biodiversity

Part III- Perspective towards biodiversity

Part IV – Biodiversity-related Laws and Policies in the Philippines

The participants of the study will be the randomly selected students in the four programs of Mambusao Satellite College enrolled for the school year 2023-2024.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1										
	March	April	May	June	July	Aug	Sept	Oct	April	May	Jun
1. Revision of proposal and gathering data											
2. Analysis of data gathered, Design, and Development of module											
3. Colloquium/defense											

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The expected output for this study is instructional material in the form of modules and audio-visual learning material that can be used in subjects like Science Technology and Society and Environmental Management in the Food Industry.
Another expected output is a pamphlet that will be available in the library.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projectsoutput)

The target beneficiaries of this study are students and faculty of Mambusao Satellite College. The learning material (module and audio-visual, pamphlet) can be used by students and faculty in Science Technology and Society and also in Environmental Management in the Food Industry. This will help students to be more knowledgeable and develop a positive perspective towards biodiversity.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 5 of 7

(18) Personnel Requirement		
Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study
Program/Project Leader (in hours):
Program/Project Members (in hours):

(19) Budgetary Requirement			
SOURCES OF EXPENSES		Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)			
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)			
- Transportation allowance			
- Meal allowance during data gathering			2,000.00
- Professional fees (laboratory analysis and other related fees)			3,000.00
- Miscellaneous (for incidental expenses)			2,000.00
- Printing			3,000.00
C. CAPITAL OUTLAY			
Total			10,000.00

(20) Literature Cited (**include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>**)

Basir, S. & Ming, K. (2019). Saving the Environment and the Dilemma of Finding the Right Solution. *The European Proceedings of Social & Behavioral Sciences: International Conference on Law, Environment and Society 2018*. <https://doi.org/10.15405/epsbs.2019.10.12>.

Coracero, E. and Malabrigo Jr., P. (2020). Carbon Storage Potential of the Tree Species along the Ultramafic Forest in Sitio Dicasalarin, Barangay Zabali, Baler, Aurora, Philippines. *AIMS Environmental Science*, 7(6), 589-601. <https://doi.org/10.3934/environsci.2020037>.

Morar, F. & Peterlicean, A. (2012). The role and importance of educating youth regarding biodiversity conservation in protected natural areas. *Procedia Economics and Finance*, 3, 1117 – 1121. [http://doi.org/10.1016/S2212-5671\(12\)00283-3](http://doi.org/10.1016/S2212-5671(12)00283-3).

Raga, J. (2013). Role of Peoples and Nations in Protecting the Natural Environment. *Acta*, 18, 142-176. <http://www.pass.va/content/dam/scienzesociali/pdf/acta18/acta18-raga.pdf>.

Šorytė, D. & Pakalniškienė, V. (2019). Why it is important to protect the environment: reasons given by children. *International Research in Geographical and Environmental Education*, pp. 228-241. <https://doi.org/10.1080/10382046.2019.1582771>.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 6 of 7

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

CURRICULUM VITAE LECEL OLANGA-BOBIER

Damayan, Sapián, Capiz
lobobier@capsu.edu.ph / 09466299997



PERSONAL INFORMATION

Name : Lecel Olanga Bobier
Date of birth : May 19, 1971
Place of birth : Damayan, Sapián, Capiz
Nationality : Filipino
Religion: : Roman Catholic
Spouse : Diven F. Bobier
Child : Dave Robin O. Bobier

EDUCATIONAL BACKGROUND

Elementary : Damayan Elementary School
Damayan, Sapián, Capiz
1978 - 1984
Secondary : Altavas National School
Altavas, Aklan
1984 - 1988
College : Bachelor of Science in Biological Science
West Visayas State University
Lapaz, Iloilo City
1988 - 1992
Graduate Studies : Master of Arts in Teaching General Science
Filamer Christian University
Roxas City
2007 – 2011
: Ph. D. in Science Education major in Biology
West Visayas State University
Lapaz, Iloilo City
2016 – 2019
Scholarship : CHED K to 12 Transition Program

ELIGIBILITY : Philippine Board Examination For Teachers

EMPLOYMENT RECORD

August 8, 1994 – November 1996 : Substitute Teacher
Panay State Polytechnic College
Sapián Campus
November 1996 to May 31, 2016 : Faculty
Capiz State University
Sapián Satellite College
June 2016 to Present : Faculty -Mambusao Satellite College


LECEL OLANGA-BOBIER